Model Architecture: gpt-oss-20b

Generated by sg_modelload

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1 Model Overview

Model name	gpt-oss-20b
Transformers class	`GptOssForCausalLM`
Path	`/models/gpt-oss-20b`
Total parameters	20,914,757,184 (20.91B)
Weight dtype	bfloat16
Checkpoint quantisation	mxfp4 (shapes below are the de-quantised ones)
Hidden size d	2,880
Decoder layers L	24
Query heads H	64
Key/value heads H_{kv}	8
Head dim d_h	64
MLP width d_{ff}	2,880
Experts E	32
Active experts per token	4
Vocabulary V	201,088
Max context length	131,072
Sliding window w	128
RMSNorm ϵ	1e-05
Tied input/output embeddings	False
Activation	SiLU
RoPE base θ	150,000

How to read this document. Section 2 writes the whole network as one composed function. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Notation

B, T	batch size, sequence length (tokens)
d	hidden size (2880)
V	vocabulary size (201,088)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 The Model as a Function

Input: token ids $x \in \{0, \ldots, V-1\}^{B \times T}$. Output: next-token logits $y \in \mathbb{R}^{B \times T \times V}$. The 24 decoder layers are written with composition powers, so a block that repeats n times appears once as $\mathcal{B}^{\circ n}$.

$$
\begin{aligned}
h_0 &= \text{Embed}(x) \\
h_L &= \Big(\big(\mathcal{B}_{\text{F}} \circ \mathcal{B}_{\text{S}} \big)^{\circ 12} \Big)(h_0) \\
y &= W_{\text{LM}}\ \text{RMSNorm}_{\text{final}}(h_L)
\end{aligned}
$$

2.1 Layer pattern

Symbol	Block type	Count	Params / block	Layer indices
$\mathcal{B}_{\text{S}}$	Sliding-window attention block	12	823,186,976	0, 2, …, 22 (every 2th)
$\mathcal{B}_{\text{F}}$	Full (global) attention block	12	823,186,976	1, 3, …, 23 (every 2th)

Language-model decoder (24 layers). layers 0–23: $[\mathcal{B}_{\mathrm{S}}, \mathcal{B}_{\mathrm{F}}]$ repeated 12×.

2.2 Block functions

$\mathcal{B}_{\mathrm{S}}$ — **Sliding-window attention block.**

$a = h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h))$	pre-norm + residual
$\mathcal{B}(h) = a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a))$	pre-norm + residual
$Q = W_Q\, u + b_Q$	$H = 64$ heads of size $d_h = 64$
$K,\ V = W_K\, u + b_K, \quad W_V\, u + b_V$	$H_{kv} = 8$ key/value heads
$Q,\ K = \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K)$	rotate by position
$S_h = s\, Q_h K_{g(h)}^{\top} + M$	mask M: causal + sliding window of $w = 128$
$P_h = \mathrm{softmax}\big([\, S_h \parallel z_h \,]\big)_{1:T}$	learned sink logit z_h soaks up attention, then is dropped
$O_h = P_h\, V_{g(h)}$	
$\mathrm{Attn}(u) = W_O\, [O_1; \ldots; O_H] + b_O$	
$\lambda = W_r\, u + b_r \in \mathbb{R}^{32}$	router score per expert
$\mathcal{I} = \mathrm{TopK}(\lambda,\ 4)$	pick 4 of 32 experts
$w = \mathrm{softmax}(\lambda_{\mathcal{I}})$	mixing weights
$[a,\, c] = W_e^{gu}\, u + b_e^{gu}$	per expert e: gate a, up c
$\mathrm{E}_e(u) = W_e^{d}\big((c+1) \odot a\, \sigma(1.702\, a)\big) + b_e^{d}$	clamped SwiGLU expert
$\mathrm{FFN}(u) = \sum_{e \in \mathcal{I}} w_e\ \mathrm{E}_e(u)$	

$\mathcal{B}_{\mathrm{F}}$ — **Full (global) attention block.**

$a = h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h))$	pre-norm + residual
$\mathcal{B}(h) = a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a))$	pre-norm + residual
$Q = W_Q\, u + b_Q$	$H = 64$ heads of size $d_h = 64$
$K,\ V = W_K\, u + b_K, \quad W_V\, u + b_V$	$H_{kv} = 8$ key/value heads
$Q,\ K = \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K)$	rotate by position
$S_h = s\, Q_h K_{g(h)}^{\top} + M$	mask M: causal
$P_h = \mathrm{softmax}\big([\, S_h \parallel z_h \,]\big)_{1:T}$	learned sink logit z_h soaks up attention, then is dropped
$O_h = P_h\, V_{g(h)}$	
$\mathrm{Attn}(u) = W_O\, [O_1; \ldots; O_H] + b_O$	
$\lambda = W_r\, u + b_r \in \mathbb{R}^{32}$	router score per expert
$\mathcal{I} = \mathrm{TopK}(\lambda,\ 4)$	pick 4 of 32 experts
$w = \mathrm{softmax}(\lambda_{\mathcal{I}})$	mixing weights
$[a,\, c] = W_e^{gu}\, u + b_e^{gu}$	per expert e: gate a, up c
$\mathrm{E}_e(u) = W_e^{d}\big((c+1) \odot a\, \sigma(1.702\, a)\big) + b_e^{d}$	clamped SwiGLU expert
$\mathrm{FFN}(u) = \sum_{e \in \mathcal{I}} w_e\ \mathrm{E}_e(u)$	

3 Forward Pass in Pseudocode

Algorithm 1 gpt-oss-20b: full forward pass

Require: token ids x of shape $B \times T$

$h \leftarrow \mathrm{Embed}(x)$ ▷ look up one vector per token: $B{\times}T{\times}2880$
for $r \leftarrow 0$ **to** 11 **do** ▷ layers 0–23: pattern of 2 repeated 12 times
  $h \leftarrow \mathcal{B}_{\mathrm{S}}(h)$ ▷ Sliding-window attention block
  $h \leftarrow \mathcal{B}_{\mathrm{F}}(h)$ ▷ Full (global) attention block
end for
$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$
$y \leftarrow h\, W_{\mathrm{LM}}^{\top}$ ▷ logits $B{\times}T{\times}201,088$
return y ▷ $\mathrm{softmax}(y)$ = next-token probabilities

Algorithm 2 Block $\mathcal{B}_{\mathrm{S}}(h)$ — Sliding-window attention block

Require: h of shape $B \times T \times 2880$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top + b_Q$ ▷ $B{\times}T{\times}64{\times}64$ (64 query heads)
$K \leftarrow uW_K^\top + b_K;\ V \leftarrow uW_V^\top + b_V$ ▷ $B{\times}T{\times}8{\times}64$ (8 key/value heads, each shared by 8 query heads)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal + only the last 128 tokens; plus a learned sink column
$O \leftarrow PV$ ▷ $B{\times}T{\times}64{\times}64$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}2880$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$\lambda \leftarrow uW_r^\top + b_r$ ▷ $B{\times}T{\times}32$ router scores
$\mathcal{I}, w \leftarrow \mathrm{TopK}(\lambda, 4),\ \mathrm{softmax}$ ▷ each token picks 4 of 32 experts
$m \leftarrow \sum_{e\in\mathcal{I}} w_e\ \mathrm{E}_e(u)$ ▷ expert: clamped SwiGLU, width 2880
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 3 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 2880$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top + b_Q$ ▷ $B{\times}T{\times}64{\times}64$ (64 query heads)
$K \leftarrow uW_K^\top + b_K;\ V \leftarrow uW_V^\top + b_V$ ▷ $B{\times}T{\times}8{\times}64$ (8 key/value heads, each shared by 8 query heads)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens; plus a learned sink column
$O \leftarrow PV$ ▷ $B{\times}T{\times}64{\times}64$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}2880$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$\lambda \leftarrow uW_r^\top + b_r$ ▷ $B{\times}T{\times}32$ router scores
$\mathcal{I}, w \leftarrow \mathrm{TopK}(\lambda, 4),\ \mathrm{softmax}$ ▷ each token picks 4 of 32 experts
$m \leftarrow \sum_{e\in\mathcal{I}} w_e\ \mathrm{E}_e(u)$ ▷ expert: clamped SwiGLU, width 2880
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 2: Weights that appear once

Weight	Type	Shape	Params
`model.embed_tokens.weight`	Embedding	201088 × 2880	579,133,440
`model.norm.weight`	GptOssRMSNorm	2880	2,880
`lm_head.weight`	Linear	201088 × 2880	579,133,440

4.2 Language-model decoder: `model.layers`

Table 3: $\mathcal{B}_{\mathrm{S}}$: Sliding-window attention block — layers 0, 2, …, 22 (every 2th); weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.sinks`	GptOssAttention	64	64
`self_attn.q_proj.weight`	Linear	4096 × 2880	11,796,480
`self_attn.q_proj.bias`	Linear	4096	4,096
`self_attn.k_proj.weight`	Linear	512 × 2880	1,474,560
`self_attn.k_proj.bias`	Linear	512	512
`self_attn.v_proj.weight`	Linear	512 × 2880	1,474,560
`self_attn.v_proj.bias`	Linear	512	512
`self_attn.o_proj.weight`	Linear	2880 × 4096	11,796,480
`self_attn.o_proj.bias`	Linear	2880	2,880
`mlp.router.weight`	GptOssTopKRouter	32 × 2880	92,160
`mlp.router.bias`	GptOssTopKRouter	32	32
`mlp.experts.gate_up_proj`	GptOssExperts	32 × 2880 × 5760	530,841,600
`mlp.experts.gate_up_proj_bias`	GptOssExperts	32 × 5760	184,320
`mlp.experts.down_proj`	GptOssExperts	32 × 2880 × 2880	265,420,800
`mlp.experts.down_proj_bias`	GptOssExperts	32 × 2880	92,160
`input_layernorm.weight`	GptOssRMSNorm	2880	2,880
`post_attention_layernorm.weight`	GptOssRMSNorm	2880	2,880
Per block			**823,186,976**
× 12 blocks			**9,878,243,712**

Table 4: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 1, 3, …, 23 (every 2th); weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.sinks`	GptOssAttention	64	64
`self_attn.q_proj.weight`	Linear	4096 × 2880	11,796,480
`self_attn.q_proj.bias`	Linear	4096	4,096
`self_attn.k_proj.weight`	Linear	512 × 2880	1,474,560
`self_attn.k_proj.bias`	Linear	512	512
`self_attn.v_proj.weight`	Linear	512 × 2880	1,474,560
`self_attn.v_proj.bias`	Linear	512	512
`self_attn.o_proj.weight`	Linear	2880 × 4096	11,796,480
`self_attn.o_proj.bias`	Linear	2880	2,880
`mlp.router.weight`	GptOssTopKRouter	32 × 2880	92,160
`mlp.router.bias`	GptOssTopKRouter	32	32
`mlp.experts.gate_up_proj`	GptOssExperts	32 × 2880 × 5760	530,841,600
`mlp.experts.gate_up_proj_bias`	GptOssExperts	32 × 5760	184,320
`mlp.experts.down_proj`	GptOssExperts	32 × 2880 × 2880	265,420,800
`mlp.experts.down_proj_bias`	GptOssExperts	32 × 2880	92,160
`input_layernorm.weight`	GptOssRMSNorm	2880	2,880
`post_attention_layernorm.weight`	GptOssRMSNorm	2880	2,880
Per block			**823,186,976**
× 12 blocks			**9,878,243,712**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).

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Layer (appears as)	What it does
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
LM head (output projection) `lm_head`	Maps each final hidden vector to one score (logit) per vocabulary token. Softmax of the logits gives the next-token probabilities. When tied, it reuses the embedding matrix.
Attention sinks `layers.self_attn`	One learned logit per head that joins the softmax as an extra "attend to nothing" slot. Lets a head put weight nowhere instead of being forced onto real tokens.
MoE router `mlp.router`	A small linear layer that scores every expert for every token. The top-k experts are run and their outputs mixed with softmax weights.
Expert gate+up projection (fused) `mlp.experts`	For each expert: one matrix producing both the gate and the up-projection (interleaved), plus a bias. Every expert is its own small MLP.
Expert down projection `mlp.experts`	For each expert: projects the gated hidden activations back to the model width (plus bias).
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.

6 Parameter Count

Table 6: Parameters by component

Component	Parameters	% of total
MLP / MoE	19,119,145,728	91.4%
Attention / token mixer	637,203,456	3.0%
Token embedding	579,133,440	2.8%
LM head	579,133,440	2.8%
Norms	141,120	0.0%
Total	**20,914,757,184**	**100.0%**

Table 7: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{S}}$ Sliding-window attention block	12	823,186,976	9,878,243,712
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	12	823,186,976	9,878,243,712
Modules outside stacks	1	–	1,158,269,760
Total			**20,914,757,184**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
lm_head.weight	Linear	201088 × 2880	bfloat16	579,133,440
model	GptOssModel	–	–	–
model.embed_tokens.weight	Embedding	201088 × 2880	bfloat16	579,133,440
model.layers.0	GptOssDecoderLayer	–	–	–
model.layers.0.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.0.mlp	GptOssMLP	–	–	–
model.layers.0.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.0.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.0.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.0.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.0.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.0.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.0.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.0.self_attn	GptOssAttention	–	–	–
model.layers.0.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.0.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.0.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.0.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.0.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.0.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.0.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.0.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.0.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.1	GptOssDecoderLayer	–	–	–
model.layers.1.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.1.mlp	GptOssMLP	–	–	–
model.layers.1.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.1.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.1.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.1.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.1.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.1.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.1.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.1.self_attn	GptOssAttention	–	–	–
model.layers.1.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.1.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.1.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.1.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.1.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.1.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.1.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.1.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.1.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.2	GptOssDecoderLayer	–	–	–
model.layers.2.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.2.mlp	GptOssMLP	–	–	–
model.layers.2.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.2.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.2.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.2.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.2.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.2.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.2.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.2.self_attn	GptOssAttention	–	–	–
model.layers.2.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.2.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.2.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.2.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.2.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.2.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.2.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.2.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.2.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.3	GptOssDecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.3.mlp	GptOssMLP	–	–	–
model.layers.3.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.3.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.3.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.3.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.3.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32

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Layer path	Type	Shape	Dtype	Params
model.layers.3.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.3.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.3.self_attn	GptOssAttention	–	–	–
model.layers.3.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.3.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.3.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.3.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.3.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.3.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.3.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.3.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.3.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.4	GptOssDecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.4.mlp	GptOssMLP	–	–	–
model.layers.4.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.4.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.4.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.4.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.4.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.4.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.4.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.4.self_attn	GptOssAttention	–	–	–
model.layers.4.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.4.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.4.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.4.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.4.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.4.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.4.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.4.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.4.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.5	GptOssDecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.5.mlp	GptOssMLP	–	–	–
model.layers.5.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.5.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.5.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.5.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.5.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.5.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.5.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.5.self_attn	GptOssAttention	–	–	–
model.layers.5.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.5.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.5.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.5.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.5.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.5.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.5.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.5.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.5.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.6	GptOssDecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.6.mlp	GptOssMLP	–	–	–
model.layers.6.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.6.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.6.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.6.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.6.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.6.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.6.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.6.self_attn	GptOssAttention	–	–	–
model.layers.6.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.6.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.6.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.6.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.6.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.6.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.6.self_attn.sinks	GptOssAttention	64	bfloat16	64

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Layer path	Type	Shape	Dtype	Params
model.layers.6.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.6.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.7	GptOssDecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.7.mlp	GptOssMLP	–	–	–
model.layers.7.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.7.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.7.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.7.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.7.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.7.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.7.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.7.self_attn	GptOssAttention	–	–	–
model.layers.7.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.7.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.7.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.7.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.7.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.7.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.7.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.7.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.7.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.8	GptOssDecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.8.mlp	GptOssMLP	–	–	–
model.layers.8.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.8.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.8.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.8.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.8.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.8.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.8.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.8.self_attn	GptOssAttention	–	–	–
model.layers.8.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.8.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.8.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.8.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.8.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.8.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.8.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.8.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.8.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.9	GptOssDecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.9.mlp	GptOssMLP	–	–	–
model.layers.9.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.9.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.9.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.9.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.9.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.9.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.9.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.9.self_attn	GptOssAttention	–	–	–
model.layers.9.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.9.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.9.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.9.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.9.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.9.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.9.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.9.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.9.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.10	GptOssDecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.10.mlp	GptOssMLP	–	–	–
model.layers.10.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.10.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.10.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.10.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.10.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.10.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160

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Layer path	Type	Shape	Dtype	Params
model.layers.10.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.10.self_attn	GptOssAttention	–	–	–
model.layers.10.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.10.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.10.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.10.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.10.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.10.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.10.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.10.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.10.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.11	GptOssDecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.11.mlp	GptOssMLP	–	–	–
model.layers.11.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.11.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.11.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.11.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.11.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.11.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.11.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.11.self_attn	GptOssAttention	–	–	–
model.layers.11.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.11.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.11.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.11.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.11.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.11.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.11.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.11.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.11.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.12	GptOssDecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.12.mlp	GptOssMLP	–	–	–
model.layers.12.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.12.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.12.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.12.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.12.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.12.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.12.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.12.self_attn	GptOssAttention	–	–	–
model.layers.12.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.12.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.12.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.12.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.12.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.12.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.12.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.12.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.12.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.13	GptOssDecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.13.mlp	GptOssMLP	–	–	–
model.layers.13.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.13.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.13.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.13.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.13.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.13.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.13.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.13.self_attn	GptOssAttention	–	–	–
model.layers.13.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.13.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.13.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.13.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.13.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.13.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.13.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.13.self_attn.v_proj.bias	Linear	512	bfloat16	512

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Layer path	Type	Shape	Dtype	Params
model.layers.13.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.14	GptOssDecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.14.mlp	GptOssMLP	–	–	–
model.layers.14.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.14.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.14.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.14.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.14.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.14.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.14.post_attention_layernorm. weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.14.self_attn	GptOssAttention	–	–	–
model.layers.14.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.14.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.14.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.14.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.14.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.14.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.14.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.14.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.14.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.15	GptOssDecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.15.mlp	GptOssMLP	–	–	–
model.layers.15.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.15.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.15.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.15.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.15.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.15.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.15.post_attention_layernorm. weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.15.self_attn	GptOssAttention	–	–	–
model.layers.15.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.15.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.15.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.15.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.15.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.15.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.15.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.15.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.15.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.16	GptOssDecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.16.mlp	GptOssMLP	–	–	–
model.layers.16.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.16.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.16.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.16.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.16.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.16.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.16.post_attention_layernorm. weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.16.self_attn	GptOssAttention	–	–	–
model.layers.16.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.16.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.16.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.16.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.16.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.16.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.16.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.16.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.16.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.17	GptOssDecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.17.mlp	GptOssMLP	–	–	–
model.layers.17.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.17.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.17.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.17.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.17.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.17.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160

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Layer path	Type	Shape	Dtype	Params
model.layers.17.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.17.self_attn	GptOssAttention	–	–	–
model.layers.17.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.17.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.17.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.17.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.17.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.17.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.17.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.17.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.17.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.18	GptOssDecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.18.mlp	GptOssMLP	–	–	–
model.layers.18.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.18.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.18.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.18.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.18.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.18.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.18.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.18.self_attn	GptOssAttention	–	–	–
model.layers.18.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.18.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.18.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.18.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.18.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.18.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.18.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.18.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.18.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.19	GptOssDecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.19.mlp	GptOssMLP	–	–	–
model.layers.19.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.19.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.19.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.19.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.19.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.19.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.19.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.19.self_attn	GptOssAttention	–	–	–
model.layers.19.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.19.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.19.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.19.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.19.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.19.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.19.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.19.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.19.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.20	GptOssDecoderLayer	–	–	–
model.layers.20.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.20.mlp	GptOssMLP	–	–	–
model.layers.20.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.20.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.20.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.20.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.20.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.20.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.20.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.20.self_attn	GptOssAttention	–	–	–
model.layers.20.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.20.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.20.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.20.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.20.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.20.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.20.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.20.self_attn.v_proj.bias	Linear	512	bfloat16	512

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Layer path	Type	Shape	Dtype	Params
model.layers.20.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.21	GptOssDecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.21.mlp	GptOssMLP	–	–	–
model.layers.21.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.21.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.21.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.21.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.21.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.21.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.21.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.21.self_attn	GptOssAttention	–	–	–
model.layers.21.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.21.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.21.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.21.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.21.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.21.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.21.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.21.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.21.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.22	GptOssDecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.22.mlp	GptOssMLP	–	–	–
model.layers.22.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.22.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.22.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.22.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.22.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.22.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.22.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.22.self_attn	GptOssAttention	–	–	–
model.layers.22.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.22.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.22.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.22.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.22.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.22.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.22.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.22.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.22.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.23	GptOssDecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.23.mlp	GptOssMLP	–	–	–
model.layers.23.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.23.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.23.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.23.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.23.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.23.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.23.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.23.self_attn	GptOssAttention	–	–	–
model.layers.23.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.23.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.23.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.23.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.23.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.23.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.23.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.23.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.23.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.norm.weight	GptOssRMSNorm	2880	bfloat16	2,880

Appendix B Per-Layer Parameter Count

Module	Params	Running total
lm_head	579,133,440	579,133,440

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Module	Params	Running total
model.embed_tokens	579,133,440	1,158,266,880
model.layers.0.input_layernorm	2,880	1,158,269,760
model.layers.0.mlp.experts	796,538,880	1,954,808,640
model.layers.0.mlp.router	92,192	1,954,900,832
model.layers.0.post_attention_layernorm	2,880	1,954,903,712
model.layers.0.self_attn.k_proj	1,475,072	1,956,378,784
model.layers.0.self_attn.o_proj	11,799,360	1,968,178,144
model.layers.0.self_attn.q_proj	11,800,576	1,979,978,720
model.layers.0.self_attn	64	1,979,978,784
model.layers.0.self_attn.v_proj	1,475,072	1,981,453,856
model.layers.1.input_layernorm	2,880	1,981,456,736
model.layers.1.mlp.experts	796,538,880	2,777,995,616
model.layers.1.mlp.router	92,192	2,778,087,808
model.layers.1.post_attention_layernorm	2,880	2,778,090,688
model.layers.1.self_attn.k_proj	1,475,072	2,779,565,760
model.layers.1.self_attn.o_proj	11,799,360	2,791,365,120
model.layers.1.self_attn.q_proj	11,800,576	2,803,165,696
model.layers.1.self_attn	64	2,803,165,760
model.layers.1.self_attn.v_proj	1,475,072	2,804,640,832
model.layers.2.input_layernorm	2,880	2,804,643,712
model.layers.2.mlp.experts	796,538,880	3,601,182,592
model.layers.2.mlp.router	92,192	3,601,274,784
model.layers.2.post_attention_layernorm	2,880	3,601,277,664
model.layers.2.self_attn.k_proj	1,475,072	3,602,752,736
model.layers.2.self_attn.o_proj	11,799,360	3,614,552,096
model.layers.2.self_attn.q_proj	11,800,576	3,626,352,672
model.layers.2.self_attn	64	3,626,352,736
model.layers.2.self_attn.v_proj	1,475,072	3,627,827,808
model.layers.3.input_layernorm	2,880	3,627,830,688
model.layers.3.mlp.experts	796,538,880	4,424,369,568
model.layers.3.mlp.router	92,192	4,424,461,760
model.layers.3.post_attention_layernorm	2,880	4,424,464,640
model.layers.3.self_attn.k_proj	1,475,072	4,425,939,712
model.layers.3.self_attn.o_proj	11,799,360	4,437,739,072
model.layers.3.self_attn.q_proj	11,800,576	4,449,539,648
model.layers.3.self_attn	64	4,449,539,712
model.layers.3.self_attn.v_proj	1,475,072	4,451,014,784
model.layers.4.input_layernorm	2,880	4,451,017,664
model.layers.4.mlp.experts	796,538,880	5,247,556,544
model.layers.4.mlp.router	92,192	5,247,648,736
model.layers.4.post_attention_layernorm	2,880	5,247,651,616
model.layers.4.self_attn.k_proj	1,475,072	5,249,126,688
model.layers.4.self_attn.o_proj	11,799,360	5,260,926,048
model.layers.4.self_attn.q_proj	11,800,576	5,272,726,624
model.layers.4.self_attn	64	5,272,726,688
model.layers.4.self_attn.v_proj	1,475,072	5,274,201,760
model.layers.5.input_layernorm	2,880	5,274,204,640
model.layers.5.mlp.experts	796,538,880	6,070,743,520
model.layers.5.mlp.router	92,192	6,070,835,712
model.layers.5.post_attention_layernorm	2,880	6,070,838,592
model.layers.5.self_attn.k_proj	1,475,072	6,072,313,664
model.layers.5.self_attn.o_proj	11,799,360	6,084,113,024
model.layers.5.self_attn.q_proj	11,800,576	6,095,913,600
model.layers.5.self_attn	64	6,095,913,664
model.layers.5.self_attn.v_proj	1,475,072	6,097,388,736
model.layers.6.input_layernorm	2,880	6,097,391,616
model.layers.6.mlp.experts	796,538,880	6,893,930,496
model.layers.6.mlp.router	92,192	6,894,022,688

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Module	Params	Running total
model.layers.6.post_attention_layernorm	2,880	6,894,025,568
model.layers.6.self_attn.k_proj	1,475,072	6,895,500,640
model.layers.6.self_attn.o_proj	11,799,360	6,907,300,000
model.layers.6.self_attn.q_proj	11,800,576	6,919,100,576
model.layers.6.self_attn	64	6,919,100,640
model.layers.6.self_attn.v_proj	1,475,072	6,920,575,712
model.layers.7.input_layernorm	2,880	6,920,578,592
model.layers.7.mlp.experts	796,538,880	7,717,117,472
model.layers.7.mlp.router	92,192	7,717,209,664
model.layers.7.post_attention_layernorm	2,880	7,717,212,544
model.layers.7.self_attn.k_proj	1,475,072	7,718,687,616
model.layers.7.self_attn.o_proj	11,799,360	7,730,486,976
model.layers.7.self_attn.q_proj	11,800,576	7,742,287,552
model.layers.7.self_attn	64	7,742,287,616
model.layers.7.self_attn.v_proj	1,475,072	7,743,762,688
model.layers.8.input_layernorm	2,880	7,743,765,568
model.layers.8.mlp.experts	796,538,880	8,540,304,448
model.layers.8.mlp.router	92,192	8,540,396,640
model.layers.8.post_attention_layernorm	2,880	8,540,399,520
model.layers.8.self_attn.k_proj	1,475,072	8,541,874,592
model.layers.8.self_attn.o_proj	11,799,360	8,553,673,952
model.layers.8.self_attn.q_proj	11,800,576	8,565,474,528
model.layers.8.self_attn	64	8,565,474,592
model.layers.8.self_attn.v_proj	1,475,072	8,566,949,664
model.layers.9.input_layernorm	2,880	8,566,952,544
model.layers.9.mlp.experts	796,538,880	9,363,491,424
model.layers.9.mlp.router	92,192	9,363,583,616
model.layers.9.post_attention_layernorm	2,880	9,363,586,496
model.layers.9.self_attn.k_proj	1,475,072	9,365,061,568
model.layers.9.self_attn.o_proj	11,799,360	9,376,860,928
model.layers.9.self_attn.q_proj	11,800,576	9,388,661,504
model.layers.9.self_attn	64	9,388,661,568
model.layers.9.self_attn.v_proj	1,475,072	9,390,136,640
model.layers.10.input_layernorm	2,880	9,390,139,520
model.layers.10.mlp.experts	796,538,880	10,186,678,400
model.layers.10.mlp.router	92,192	10,186,770,592
model.layers.10.post_attention_layernorm	2,880	10,186,773,472
model.layers.10.self_attn.k_proj	1,475,072	10,188,248,544
model.layers.10.self_attn.o_proj	11,799,360	10,200,047,904
model.layers.10.self_attn.q_proj	11,800,576	10,211,848,480
model.layers.10.self_attn	64	10,211,848,544
model.layers.10.self_attn.v_proj	1,475,072	10,213,323,616
model.layers.11.input_layernorm	2,880	10,213,326,496
model.layers.11.mlp.experts	796,538,880	11,009,865,376
model.layers.11.mlp.router	92,192	11,009,957,568
model.layers.11.post_attention_layernorm	2,880	11,009,960,448
model.layers.11.self_attn.k_proj	1,475,072	11,011,435,520
model.layers.11.self_attn.o_proj	11,799,360	11,023,234,880
model.layers.11.self_attn.q_proj	11,800,576	11,035,035,456
model.layers.11.self_attn	64	11,035,035,520
model.layers.11.self_attn.v_proj	1,475,072	11,036,510,592
model.layers.12.input_layernorm	2,880	11,036,513,472
model.layers.12.mlp.experts	796,538,880	11,833,052,352
model.layers.12.mlp.router	92,192	11,833,144,544
model.layers.12.post_attention_layernorm	2,880	11,833,147,424
model.layers.12.self_attn.k_proj	1,475,072	11,834,622,496
model.layers.12.self_attn.o_proj	11,799,360	11,846,421,856
model.layers.12.self_attn.q_proj	11,800,576	11,858,222,432

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Module	Params	Running total
model.layers.12.self_attn	64	11,858,222,496
model.layers.12.self_attn.v_proj	1,475,072	11,859,697,568
model.layers.13.input_layernorm	2,880	11,859,700,448
model.layers.13.mlp.experts	796,538,880	12,656,239,328
model.layers.13.mlp.router	92,192	12,656,331,520
model.layers.13.post_attention_layernorm	2,880	12,656,334,400
model.layers.13.self_attn.k_proj	1,475,072	12,657,809,472
model.layers.13.self_attn.o_proj	11,799,360	12,669,608,832
model.layers.13.self_attn.q_proj	11,800,576	12,681,409,408
model.layers.13.self_attn	64	12,681,409,472
model.layers.13.self_attn.v_proj	1,475,072	12,682,884,544
model.layers.14.input_layernorm	2,880	12,682,887,424
model.layers.14.mlp.experts	796,538,880	13,479,426,304
model.layers.14.mlp.router	92,192	13,479,518,496
model.layers.14.post_attention_layernorm	2,880	13,479,521,376
model.layers.14.self_attn.k_proj	1,475,072	13,480,996,448
model.layers.14.self_attn.o_proj	11,799,360	13,492,795,808
model.layers.14.self_attn.q_proj	11,800,576	13,504,596,384
model.layers.14.self_attn	64	13,504,596,448
model.layers.14.self_attn.v_proj	1,475,072	13,506,071,520
model.layers.15.input_layernorm	2,880	13,506,074,400
model.layers.15.mlp.experts	796,538,880	14,302,613,280
model.layers.15.mlp.router	92,192	14,302,705,472
model.layers.15.post_attention_layernorm	2,880	14,302,708,352
model.layers.15.self_attn.k_proj	1,475,072	14,304,183,424
model.layers.15.self_attn.o_proj	11,799,360	14,315,982,784
model.layers.15.self_attn.q_proj	11,800,576	14,327,783,360
model.layers.15.self_attn	64	14,327,783,424
model.layers.15.self_attn.v_proj	1,475,072	14,329,258,496
model.layers.16.input_layernorm	2,880	14,329,261,376
model.layers.16.mlp.experts	796,538,880	15,125,800,256
model.layers.16.mlp.router	92,192	15,125,892,448
model.layers.16.post_attention_layernorm	2,880	15,125,895,328
model.layers.16.self_attn.k_proj	1,475,072	15,127,370,400
model.layers.16.self_attn.o_proj	11,799,360	15,139,169,760
model.layers.16.self_attn.q_proj	11,800,576	15,150,970,336
model.layers.16.self_attn	64	15,150,970,400
model.layers.16.self_attn.v_proj	1,475,072	15,152,445,472
model.layers.17.input_layernorm	2,880	15,152,448,352
model.layers.17.mlp.experts	796,538,880	15,948,987,232
model.layers.17.mlp.router	92,192	15,949,079,424
model.layers.17.post_attention_layernorm	2,880	15,949,082,304
model.layers.17.self_attn.k_proj	1,475,072	15,950,557,376
model.layers.17.self_attn.o_proj	11,799,360	15,962,356,736
model.layers.17.self_attn.q_proj	11,800,576	15,974,157,312
model.layers.17.self_attn	64	15,974,157,376
model.layers.17.self_attn.v_proj	1,475,072	15,975,632,448
model.layers.18.input_layernorm	2,880	15,975,635,328
model.layers.18.mlp.experts	796,538,880	16,772,174,208
model.layers.18.mlp.router	92,192	16,772,266,400
model.layers.18.post_attention_layernorm	2,880	16,772,269,280
model.layers.18.self_attn.k_proj	1,475,072	16,773,744,352
model.layers.18.self_attn.o_proj	11,799,360	16,785,543,712
model.layers.18.self_attn.q_proj	11,800,576	16,797,344,288
model.layers.18.self_attn	64	16,797,344,352
model.layers.18.self_attn.v_proj	1,475,072	16,798,819,424
model.layers.19.input_layernorm	2,880	16,798,822,304
model.layers.19.mlp.experts	796,538,880	17,595,361,184

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Module	Params	Running total
model.layers.19.mlp.router	92,192	17,595,453,376
model.layers.19.post_attention_layernorm	2,880	17,595,456,256
model.layers.19.self_attn.k_proj	1,475,072	17,596,931,328
model.layers.19.self_attn.o_proj	11,799,360	17,608,730,688
model.layers.19.self_attn.q_proj	11,800,576	17,620,531,264
model.layers.19.self_attn	64	17,620,531,328
model.layers.19.self_attn.v_proj	1,475,072	17,622,006,400
model.layers.20.input_layernorm	2,880	17,622,009,280
model.layers.20.mlp.experts	796,538,880	18,418,548,160
model.layers.20.mlp.router	92,192	18,418,640,352
model.layers.20.post_attention_layernorm	2,880	18,418,643,232
model.layers.20.self_attn.k_proj	1,475,072	18,420,118,304
model.layers.20.self_attn.o_proj	11,799,360	18,431,917,664
model.layers.20.self_attn.q_proj	11,800,576	18,443,718,240
model.layers.20.self_attn	64	18,443,718,304
model.layers.20.self_attn.v_proj	1,475,072	18,445,193,376
model.layers.21.input_layernorm	2,880	18,445,196,256
model.layers.21.mlp.experts	796,538,880	19,241,735,136
model.layers.21.mlp.router	92,192	19,241,827,328
model.layers.21.post_attention_layernorm	2,880	19,241,830,208
model.layers.21.self_attn.k_proj	1,475,072	19,243,305,280
model.layers.21.self_attn.o_proj	11,799,360	19,255,104,640
model.layers.21.self_attn.q_proj	11,800,576	19,266,905,216
model.layers.21.self_attn	64	19,266,905,280
model.layers.21.self_attn.v_proj	1,475,072	19,268,380,352
model.layers.22.input_layernorm	2,880	19,268,383,232
model.layers.22.mlp.experts	796,538,880	20,064,922,112
model.layers.22.mlp.router	92,192	20,065,014,304
model.layers.22.post_attention_layernorm	2,880	20,065,017,184
model.layers.22.self_attn.k_proj	1,475,072	20,066,492,256
model.layers.22.self_attn.o_proj	11,799,360	20,078,291,616
model.layers.22.self_attn.q_proj	11,800,576	20,090,092,192
model.layers.22.self_attn	64	20,090,092,256
model.layers.22.self_attn.v_proj	1,475,072	20,091,567,328
model.layers.23.input_layernorm	2,880	20,091,570,208
model.layers.23.mlp.experts	796,538,880	20,888,109,088
model.layers.23.mlp.router	92,192	20,888,201,280
model.layers.23.post_attention_layernorm	2,880	20,888,204,160
model.layers.23.self_attn.k_proj	1,475,072	20,889,679,232
model.layers.23.self_attn.o_proj	11,799,360	20,901,478,592
model.layers.23.self_attn.q_proj	11,800,576	20,913,279,168
model.layers.23.self_attn	64	20,913,279,232
model.layers.23.self_attn.v_proj	1,475,072	20,914,754,304
model.norm	2,880	20,914,757,184